

INSTRUCTIONS TO THE CANDIDATES

- ❖ Do not use sketch pens/ high –lighter pens/ color pencils etc., in underlining / highlighting certain points in the answer or while drawing sketches etc.,
- ❖ Check the Answer booklet thoroughly. The defective answer booklet may be returned to the invigilator before starting the examination.
- ❖ Ensure that the OMR Barcode sheet is properly secured to the answer booklet.
- ❖ Do not write on or tamper the Barcode. If you do so, your booklet will not be evaluated.
- ❖ The candidate is prohibited from writing his/ her NAME / ADDRESS/ PHONE NUMBER / EMAIL ID / Religion or any other matter on any part of the answer booklet / drawing sheet/ graph sheet/ addressing the examiner in any manner whatsoever in the answer booklet. The candidate is not supposed to tear a page in full or part of the booklet.

Non-adherence with this instruction shall result in candidate being booked under malpractice.

- ❖ Answer should be written on both sides of the paper. Candidate should write in all 25 lines in each page. It is not necessary to begin each answer in a fresh page.
- ❖ Candidate should write only question number in the margin.
- ❖ Candidate should return the answer booklet to the invigilator before leaving the examination hall.
- ❖ **ADDITIONAL SUPPLEMENT OF ANSWER SHEETS WILL NOT BE ISSUED.**

INSTRUCTIONS TO THE EXAMINERS

- ❖ Invigilators should ensure that the above instructions are strictly followed by the candidates.
- ❖ After verifying the particulars on the OMR sheet, invigilator should sign in the box specified.
- ❖ Do not crumple or cut the sides of the sheets.
- ❖ Do not write, mark or damage the timing tracks or Barcode.

1
A

Fifth Normal Form (5NF)

The 5NF is one of the many normal forms.

- All the values should be atomic
- No Partial dependency
- No Transitive dependency
- All determinate should be candidate key
- No more than 1 Multivalued dependency
- Lossy decomposition

It follows Lossy decomposition.

Decomposition is the ~~dividing~~ dividing of tables.

In Lossy decomposition when the tables are reconstructed the original table will not reappear.

B Hashing

Hashing is a way of storing data. The dataset is passed through a hash function and the data is stored accordingly.

We here are using Hashing as index for databases.

It helps the user to access the database quickly.

It helps to keep track of databases.

C File handling.

It is the storing and organisation of file.

There are 5 types:

Sequential

There are 5 types.

- Sequential.
- Heap
- B + tree
- Hash
- Clustered.

Sequential. → Every thing is one after another.

Heap → It is added at last of file.

B + tree → It is stored in tree form.

Hash → Passed through a hash function.

Clustered → Repetition of data in same file.

D

Transactions.

Concurrently is when multiple people access it at the same time.

The issues we can face are

- Overloading
- Time delay

F

Roll back → Go back to a save point.

Resource starvation → When there is a high demand for that resource that it is unavailable.

4

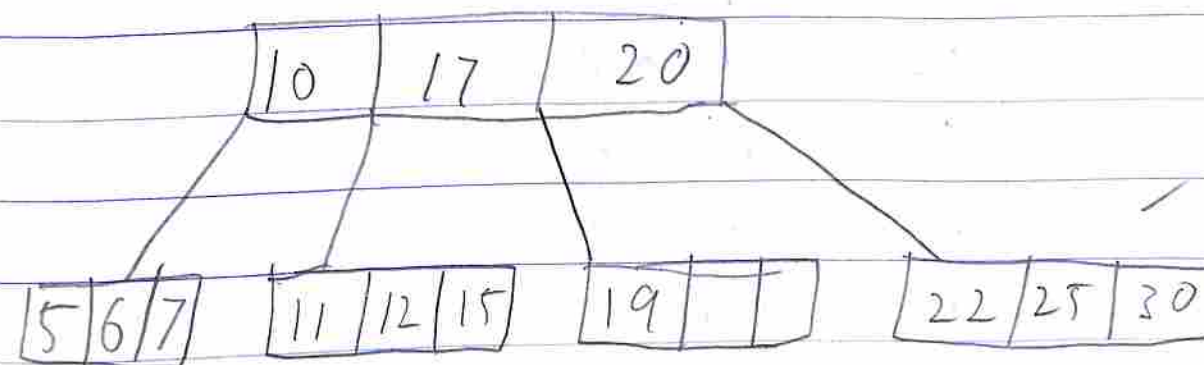
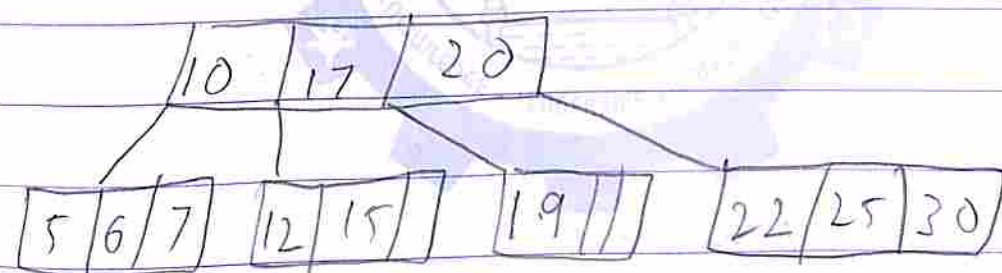
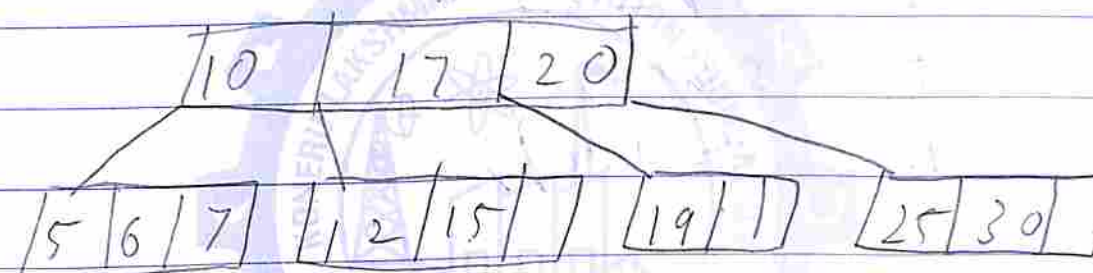
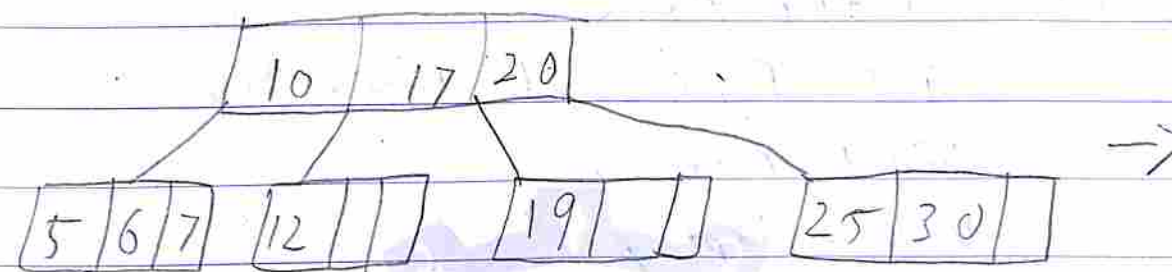
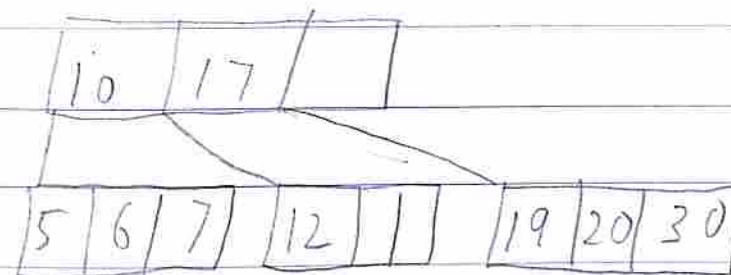
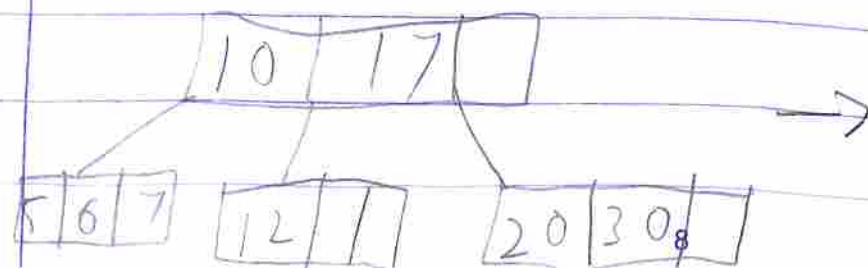
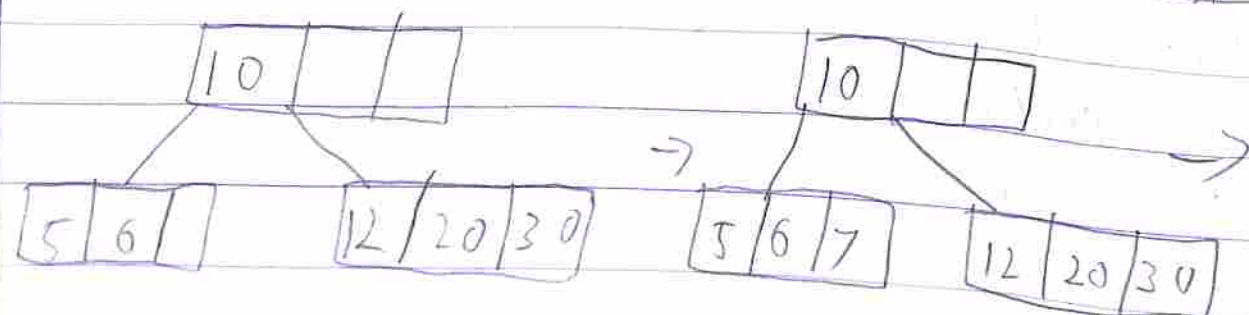
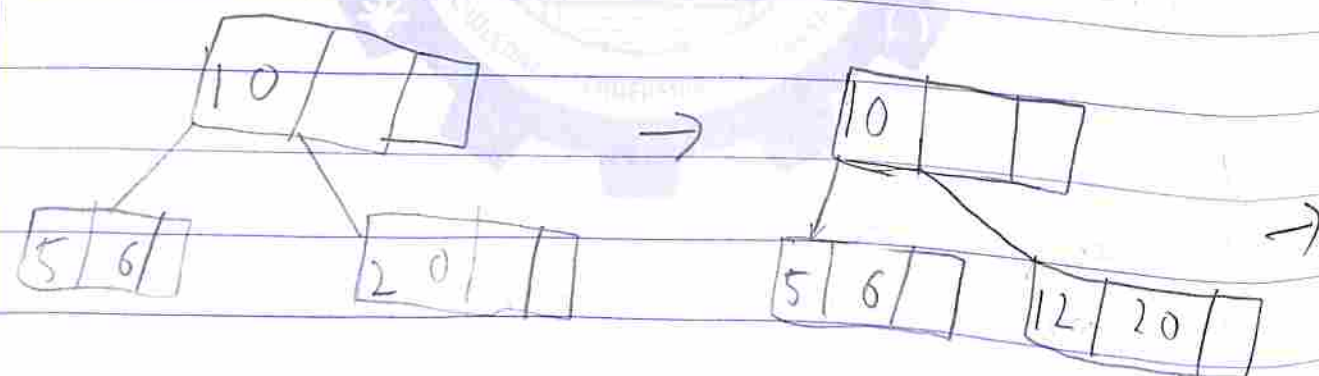
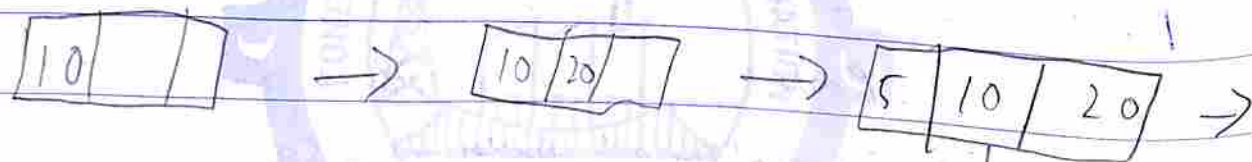
A

B+ Tree

It is a way of storing data in a tree form.

Order = 3

10, 20, 5, 6, 12, 30, 7, 17, 19, 25, 15, 22, 11



B

Create table employee (

Int Employee-ID Primary Key,
 Varchar(30) name,
 Varchar(20) Dept.,
 Int Dept-id,
 Int Salary,
 Data Hire-Date);

2

C

Time	Transaction
T1	Read (X)
T2	Write (X)
T3	Read (X)
T1	Write (X)
T2	Commit
T3	Write (X)
T1	Commit
T3	Commit

T2 did not read their account,

This is the conflict.

With this console log we were able to find this error.

A

SNo	Subject	Activity
100	Oracle	Chess
100	Oracle	Tennis
101	Java	Tennis
101	Java	Swimming
101	Java	Chess
EmpID	Project	Skill
001	Project A	Java
001	A	SQL
001	B	Java
002	A	Py
002	B	Py

SNO	Subject
100	Oracle
101	Java

SNO	Activity
100	chess
100	Tennis
101	Tennis
101	Swimming
101	chess

EmpID	Proje
001	A
001	B
002	A
002	B

Emp	SKILL
001	Java
001	SQL
001	Java
002	Py
002	Py

5.

A.

Time	Transaction
T ₁	Read(A)
T ₂	Write(B)
T ₃	Read(C)
T ₁	Write(D)
T ₂	commit
T ₃	Write(E)
T ₁	commit

T ₁	T ₂	T ₃
Read(A)		
	Write(B)	
		Read(C)
Write(D)		
	commit	
		Write(E)
commit		

B.

$$X = 600$$

$$Y = 900$$

$$M = 215$$

$$N = 65$$

Schedule A		Schedule B	
T1	T2	T1	T2
Read (X)		Read (X)	
$X := X - N$		$X := X - N$	
	Read (X)	Write (X)	
	$X := X + M$	Read (Y)	
Write (X)		$Y := Y + M$	
Read (Y)		Write (Y)	
	Write (X)		Read (X)
$Y := Y + N$			$X := X + M$
Write (Y)			Write (X)

↑

$$Y = 965$$

$$X = 750$$

The best optimum Schedule is

(B)

